AUSTO-AUTOMOBILE

Project Report

Project Details:

Project Name	Austo Automobile – Data Analysis
Original Author	Reshma B
Project Description	Auto Motor Company is a leading car manufacturer specializing in SUV, Sedan and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used.
Date	19-05-2024

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1. Project Background

Auto Motor Company is a leading car manufacturer specializing in SUV, Sedan and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used.

2. Objective

To derive practical insights from the data accumulated in order to obtain a comprehensive understanding on the demand of customers which will help them in enhancing their customer experience.

Our primary focus will be on addressing the following issues:

- Variables that influence sales of the automobile
- Factors influencing automobile sales and the respective causes
- Insights into managing demand of customers
- Methods to boost sales of each category of automobiles

3. Data description

VARIABLE	DESCRIPTION
Age	Age of the individual in years
Gender	Gender of the individual (Male/Female)
Profession	Occupation or profession of the individual
Marital_status	Marital status of the individual (Married/Single)
Education	Educational qualification of the individual (Graduate/Post Graduate)
No_of_Dependents	Number of dependents that the individual supports financially
Personal_loan	Individual taken personal loan (Yes/No)
House_loan	Individual taken housing loan (Yes/No)
Partner_working	Individual's partner employed (Yes/No)
Salary	Individual's salary or income
Partner_salary	Salary or income of the individual's partner, if applicable
Total_salary	Total combined salary of the individual and their partner (if applicable)
Price	Price of a product or service
Make	Type of automobile

4. Exploratory Data Analysis

4.1 Data Structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

OBSERVATION	VARIABLES
1581	14

4.2 Data Types

- The data set contains fourteen variables indexed from 0-13.
- The data set consists of six numerical columns, including one variable of float data type and five variables of integer data type and eight categorical columns.
- Gender and Partner_salary columns comprise null values.

4.3 Missing Values

- Gender and Partner_salary columns comprise 53 and 106 missing values respectively.

4.4 Duplicate Values

- No duplicates were found in the data set.

4.5 Data Irregularities

- Gender column had spell errors which were rectified.

4.6 Statistical Summary

The following observations were made:

- **Age:** On an average, the customers were 32 years and 75% of customers were below 38 years.
- **Gender:** Majority of customers were males, accounting to 75%, followed by females at 17% and null values at 3.3%.
- **Profession:** Salaried customers make up 56%, indicating a slightly lower rate of automobile purchases among business involves customers. However, percentage difference is only 6%.
- **Marital_status:** 90 %of the customers were married, implying that married people are more likely to purchase four-wheelers.
- **Education:** Postgraduates comprise 62% of the customer base.
- **No_of_Dependents:** The average number of dependents is 2, maximum being 4.
- **Personal_loan:** The majority of customers, totalling 985 had personal loans.
- **House_loan:** House loans were held by only 33% of customers.
- **Salary:** Half of the customers had salary below 59500 and 75% of the customers had salary below 71800.
- **Partner_Salary:** 75% of partners had salary below 38300, but however the maximum value accounts to 80500, implying the presence of outliers.
- **Total_salary:** The observed total salary ranged from a minimum of 30000 to maximum of 171000.
- **Price:** The average price of the automobiles was approximately 35000 dollars.
- **Make:** Among the 3 models, Sedan emerges as the best-selling model.

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4.7 Count and Percentage of Categorical Variable in the Data Set

GENDER	COUNT	PERCENTAGE
Male	1199	78
Female	329	22

PROFESSION	COUNT	PERCENTAGE
Salaried	896	57
Business	685	43

MARITAL STATUS	COUNT	PERCENTAGE
Yes	1443	91
No	138	8

HOUSE LOAN	COUNT	PERCENTAGE
Yes	1054	66
No	527	34

PERSONAL LOAN	COUNT	PERCENTAGE
Yes	792	51
No	789	49

PARTNER WORKING	COUNT	PERCENTAGE
Yes	868	54
No	713	46

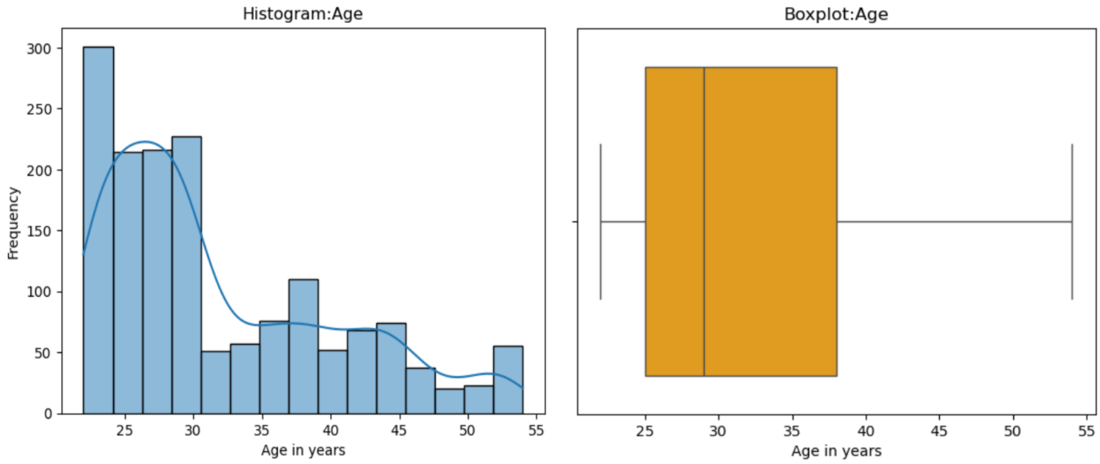
EDUCATION	COUNT	PERCENTAGE
Post graduate	985	63
Graduate	596	37

MAKE	COUNT	PERCENTAGE
Sedan	702	45
Hatchback	582	36
SUV	297	19

4.8 Univariate Analysis

4.8.1 Numerical variables

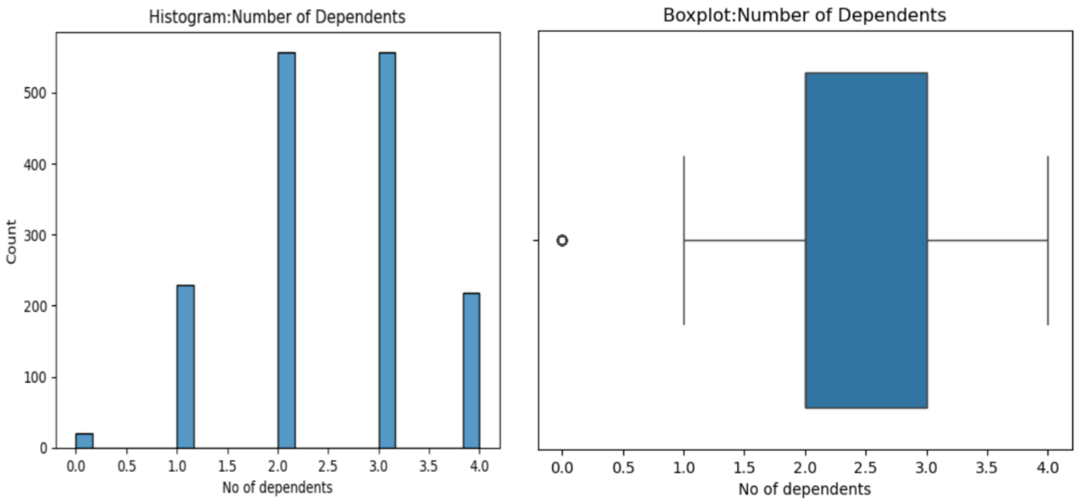
4.8.1.1 Age



Observations:

- The distribution is a highly right skewed distribution.
- The highest number of customers fall between 24-30 years of age.
- On an average, 50% of the customers are below 30 years of age.

4.8.1.2 Number of Dependents



Observations:

- It is a multimodal distribution, where the number of dependents range from 0-4.
- The median is 2, which coincides with the first quartile (Q1).
- The number of customers having 2-3 dependents constitute 35%.
- A family having zero dependent is acceptable.
- 98% of customers have dependents.

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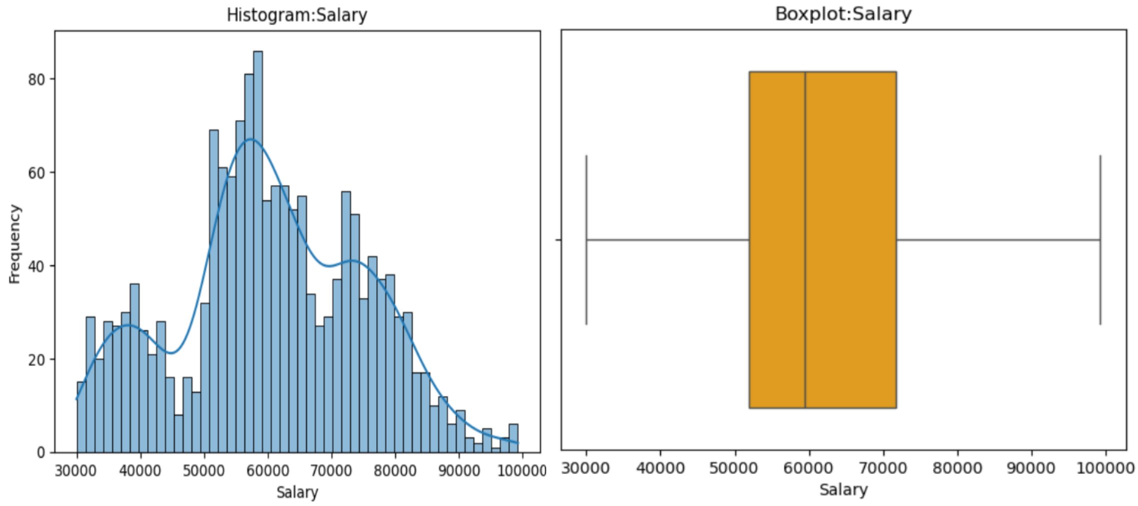
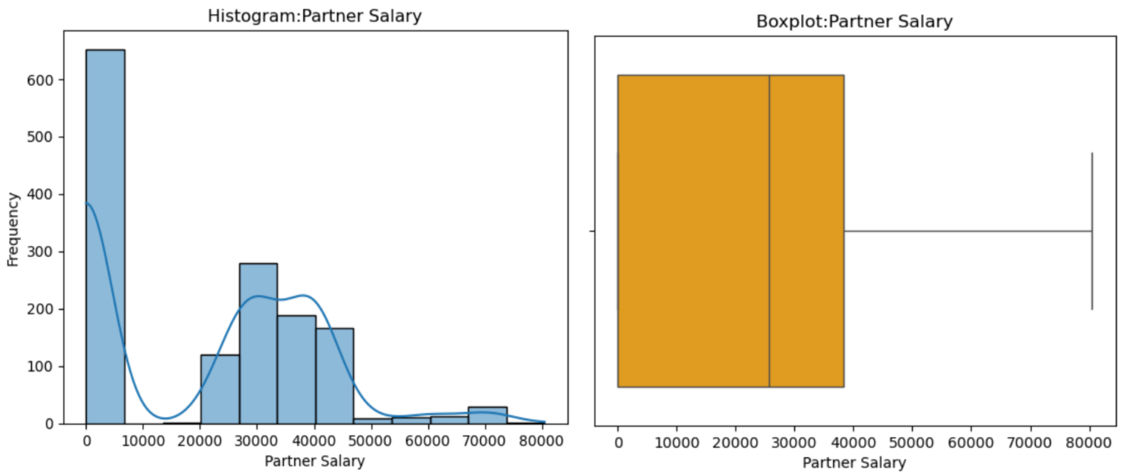
4.8.1.3 Salary



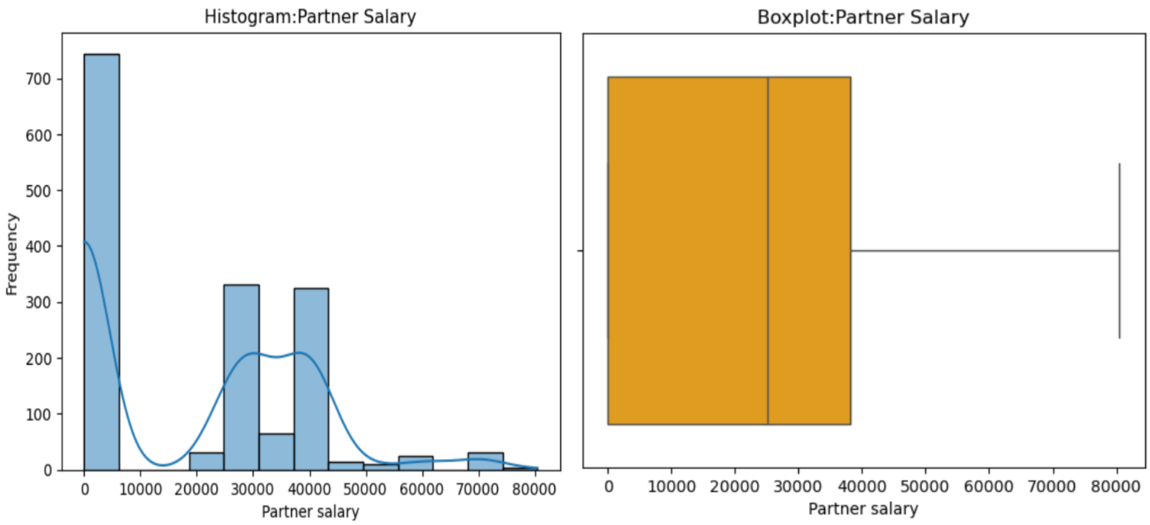
Observations:

- The average salary of the customers is less $60000.
- 75% of the customers have a salary below $72000.
- Majority of the customers earn between $50000-$70000.
- It is slightly right-skewed distribution.

4.8.1.4 Partner salary



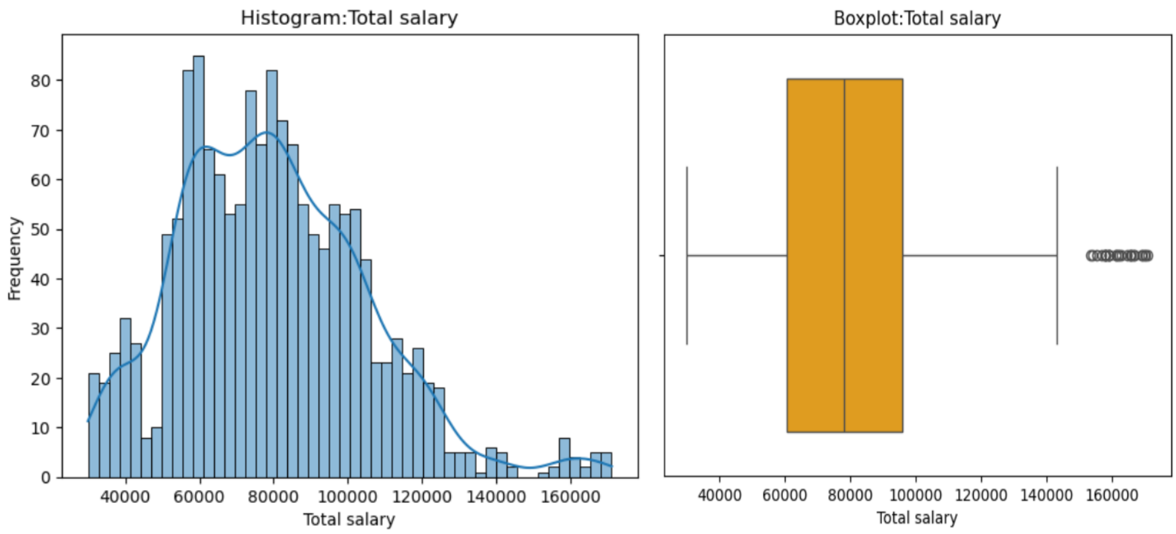
4.8.1.5 Treating missing values in Partner values



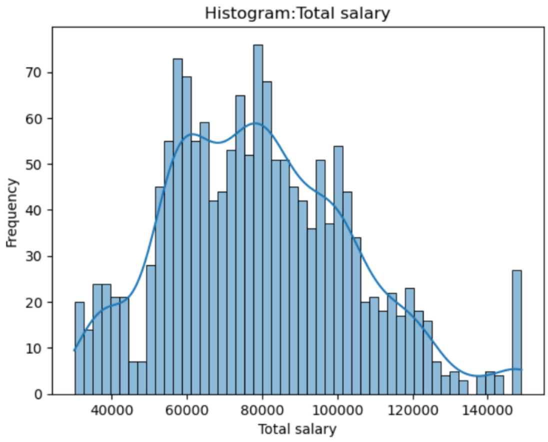
Observations:

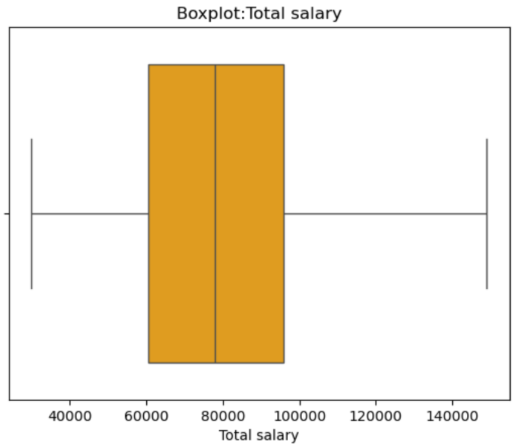
- It is a left-skewed distribution even after treating the missing values.
- Partner salary shows bimodal distribution, where the partners having salary range \$25000-\$30000 and \$38000-\$42000 are high in number.
- 75% of the partners earn less than 40000.

4.8.1.6 Total salary



4.8.1.7 Treating outliers in Total salary

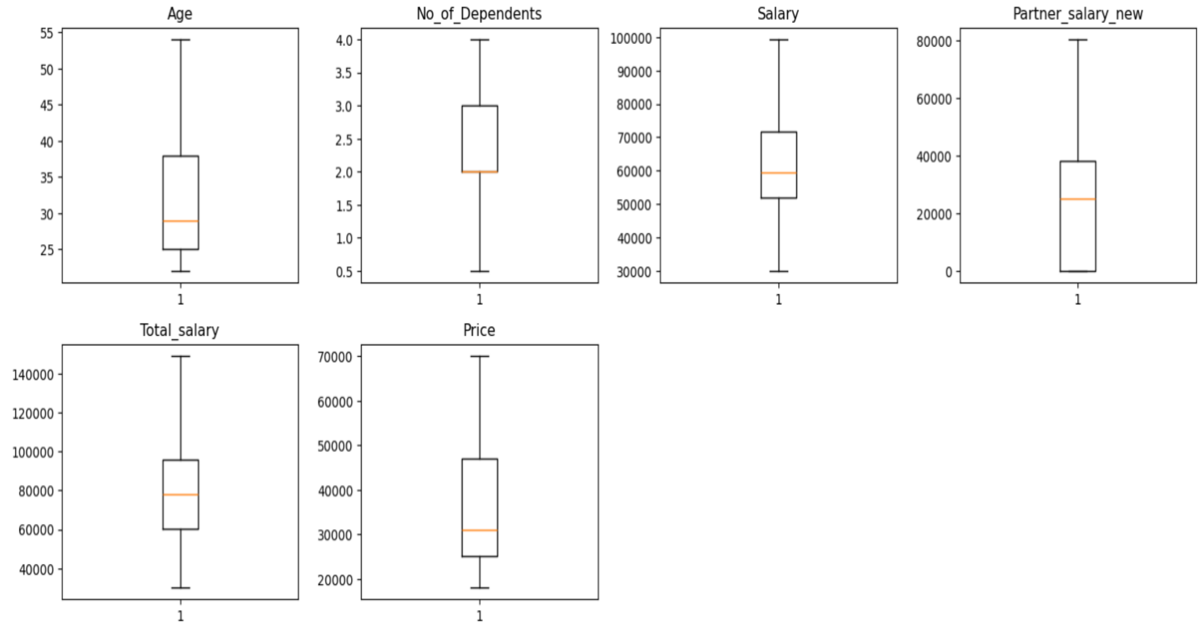




Observations:

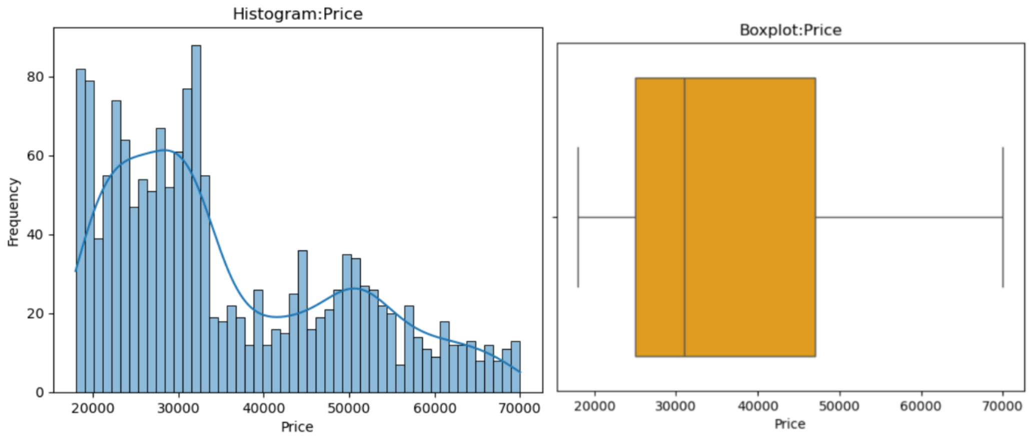
- The customers earning $60000-$100000 are high in number.
- Around 34 customers earn more than $140000, which constitutes the outliers and has been treated. Both the individual and the partner have high earnings.
- It is slightly right-skewed distribution.

4.8.1.8 Treated outliers



All the outliers have been treated.

4.8.1.9 Price

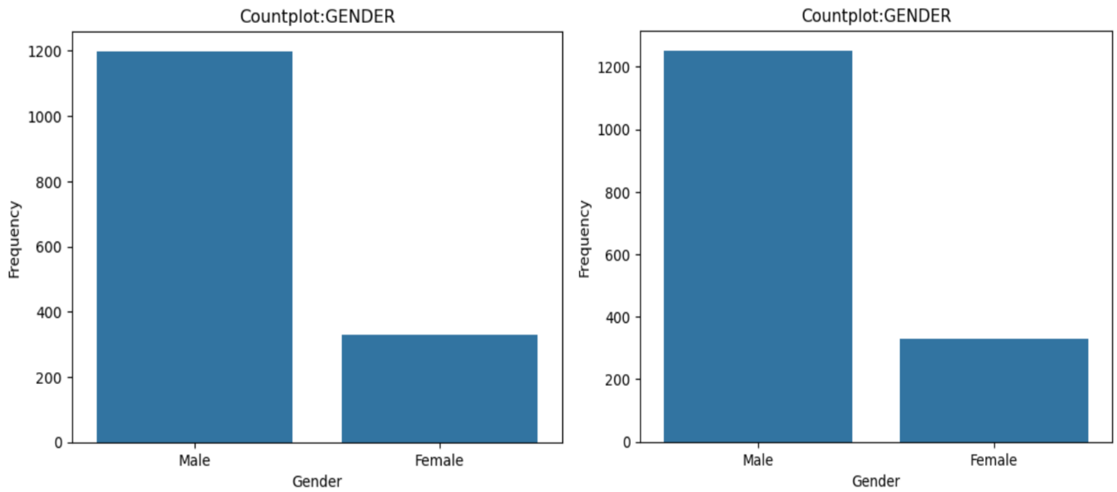


Observations:

- The distribution is a highly right-skewed distribution.
- 50 percentage of the price of the automobile is below 30000 dollars.
- Only 25% of the automobiles range between 47000-70000 dollars.
- There are no outliers in the distribution.

4.8.2 Categorical variables

4.8.2.1 Gender



GENDER	COUNT	PERCENTAGE
Male	1199	78
Female	329	22

Observations:

- The percentage of the male customers have increased from 75% to 78% after treating the missing values by replacing the mode as this is a categorical variable.
- The distribution of the Gender still remains the same.

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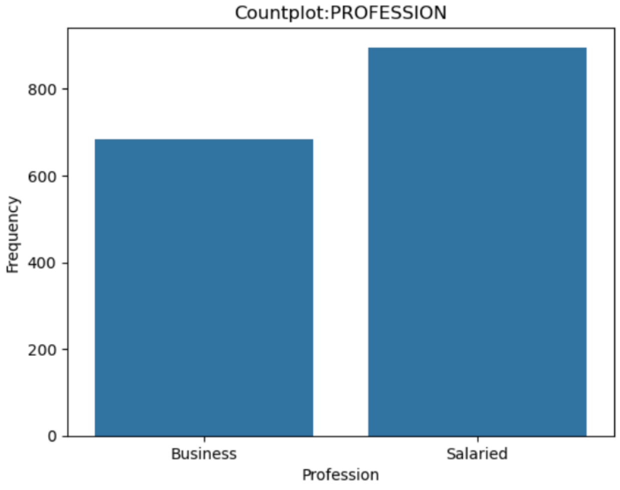
4.8.2.2 Profession

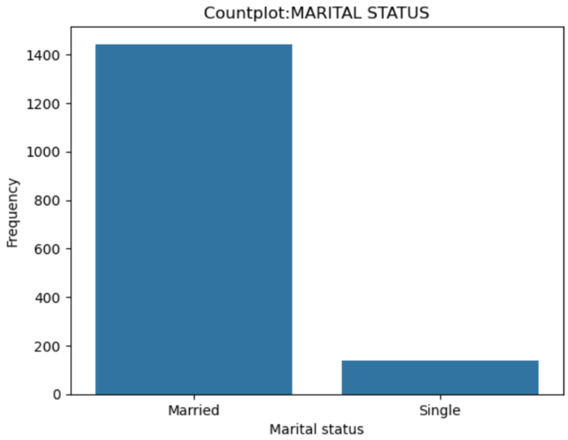


PROFESSION	COUNT	PERCENTAGE
Salaried	896	57
Business	685	43

Observations:

- Salaried customers are slightly higher than the Business customers.

4.8.2.3 Marital status

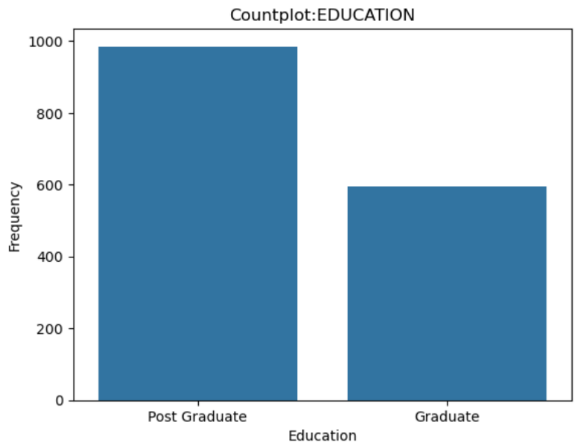


MARITAL STATUS	COUNT	PERCENTAGE
Yes	1443	91
No	138	8

Observations:

- 91 percentage of the customers are married.

4.8.2.4 Education

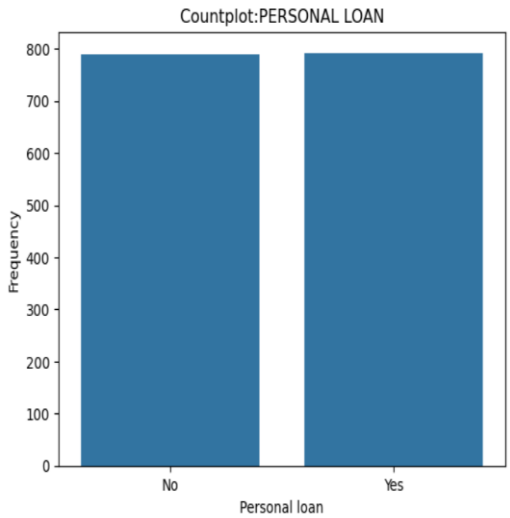


EDUCATION	COUNT	PERCENTAGE
Post graduate	985	63
Graduate	596	37

Observations:

- The percentage of post graduates are higher by approximately 40 percentage.

4.8.2.5 Personal loan



PERSONAL LOAN	COUNT	PERCENTAGE
Yes	792	51
No	789	49

Observations:

- There is equal distribution seen in both 'Yes' and 'No' parameters of the personal loan variable.
- The impact of personal loan on the purchase of automobiles can be defined by further analysis.

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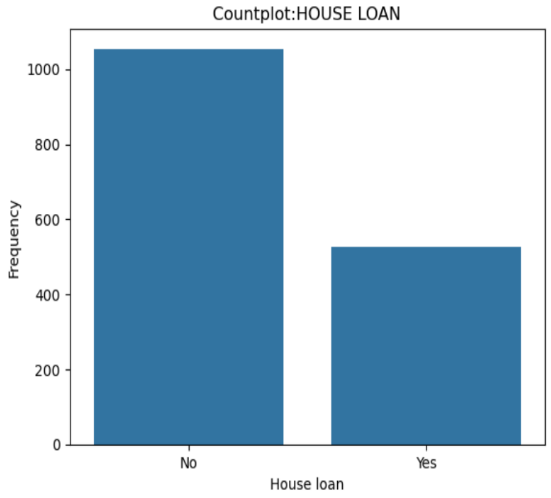
4.8.2.6 House loan

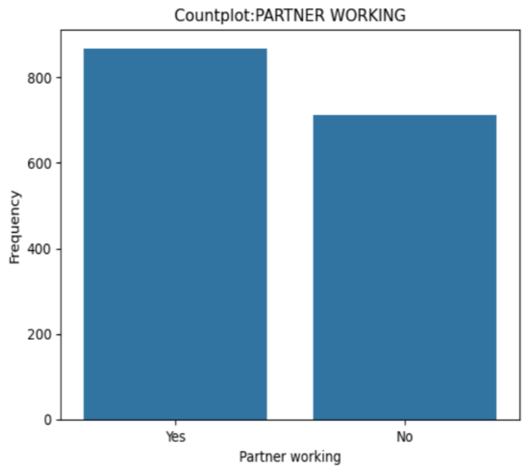


HOUSE LOAN	COUNT	PERCENTAGE
Yes	1054	66
No	527	34

Observations:

- Most of the customers do not have a house loan.
- Only 34% of the customers have house loan.

4.8.2.7 Partner working

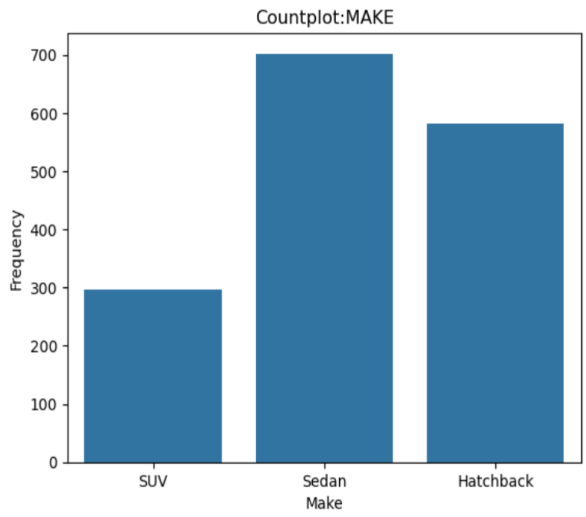


PARTNER WORKING	COUNT	PERCENTAGE
Yes	868	54
No	713	46

Observations:

- The number of partners working are slightly higher than the non-working partners.

4.8.2.8 Make of the automobile



MAKE	COUNT	PERCENTAGE
Sedan	702	45
Hatchback	582	36
SUV	297	19

Observations:

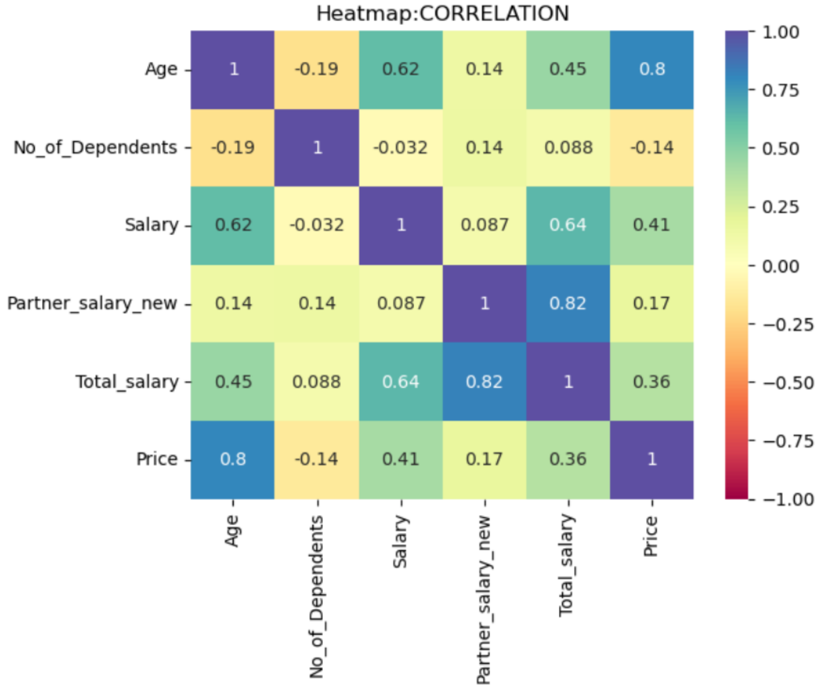
- Sedan is the top selling model, followed by hatchback and SUV.
- SUV is the lowest selling model, approximately only 20%.

4.8.3 Insights of Univariate Analysis

- Age has been an important factor, as customers falling below 30 years of age tend to purchase automobiles.
- 98% percentage of customers have dependents, hence there is a need for a family of 3 or 4 to purchase 4 wheelers.
- As there are different models of the automobiles, individuals with all range of salaries tend to purchase automobiles as per their earnings. The contribution of working partners is not particularly substantial. The distribution is wide. Hence it is necessary to consider different target customers for selling different variants of automobiles.
- Though price of the automobiles ranges from $20000-$70000, half of the customers tend to purchase cars below $30000.
- Married men constitute the majority in purchasing automobiles.
- Individuals without house loans prefer buying automobiles, however individuals with personal loans buying automobiles constitute 50%. As house loans require significant funds, there might be concerns about allocation additional money towards purchasing automobiles.

4.9 Bivariate Analysis

4.9.1 Correlation Between All Numerical Variables



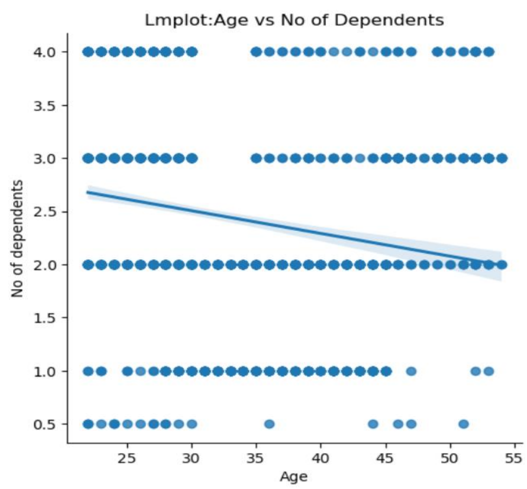
	Age	No_of_Dependents	Salary	Partner_salary_new	Total_salary	Price
Age	1	-0.19	0.62	0.14	0.45	0.8
No_of_Dependents	-0.19	1	-0.032	0.14	0.088	-0.14
Salary	0.62	-0.032	1	0.087	0.64	0.41
Partner_salary_new	0.14	0.14	0.087	1	0.82	0.17
Total_salary	0.45	0.088	0.64	0.82	1	0.36
Price	0.8	-0.14	0.41	0.17	0.36	1

Observations:

- High positive correlation can be observed between age and price (0.8), age and salary (0.62), total salary and partner salary (0.82), salary and price (0.41) and total salary and price (0.37).
- Negative correlation is observed between age and number of dependents (-0.19).

4.9.2 Relationship between all numerical variables

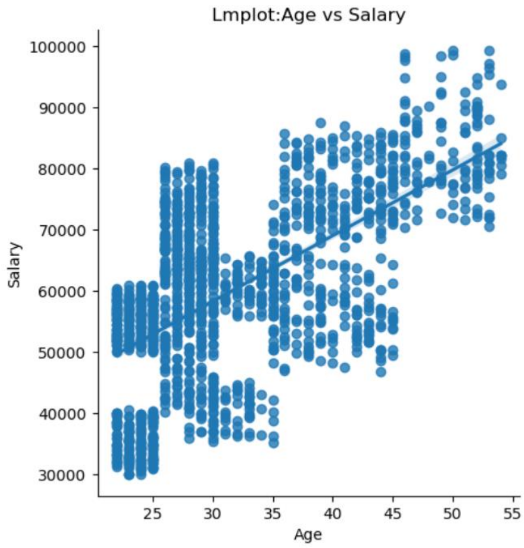
4.9.2.1 What impact does age have on the number of dependents?



Observations:

- As age increases, the number of dependents decreases, which suggests most of the dependents could be elders.
- Negative correlation is observed between age and number of dependents.

4.9.2.2 How does one's age impact their salary?



Observations:

- As age increases, salary also increases which suggests with experience, salary increases.
- Positive correlation (0.61) is observed between age and salary.

4.9.2.3 Is there any correlation between an individual's age and their partner's salary?

Observations:

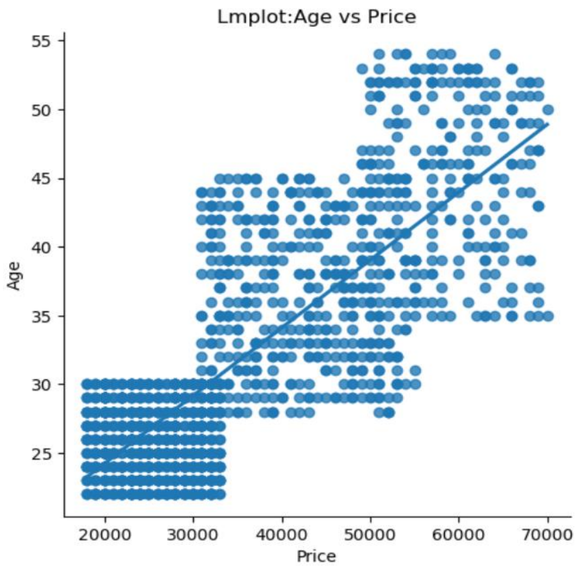
- As age increases, partner salary slightly increases. As the age of partner increases with the age of customer, it justifies salary increases with experience.
- Positive correlation is observed between age and partner.

4.9.2.4 How does the age of an individual affect both their salary and the salary of their partner?

Observations:

- As age increases, total salary also increases which suggests with experience, salary increases.
- Positive correlation (0.45) is observed between age and total salary.

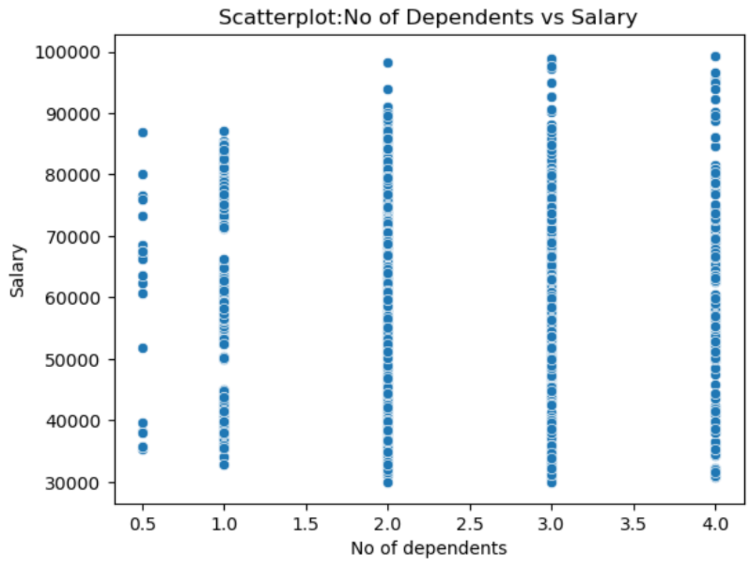
4.9.2.5 How does an individual's age influence the price of the car they buy?



Observations:

- As age increases, the price of the automobile purchased also increases. As age increases and salary increases, thereby customers afford to purchase higher priced automobiles.
- Positive correlation (0.8) is observed between age and price.

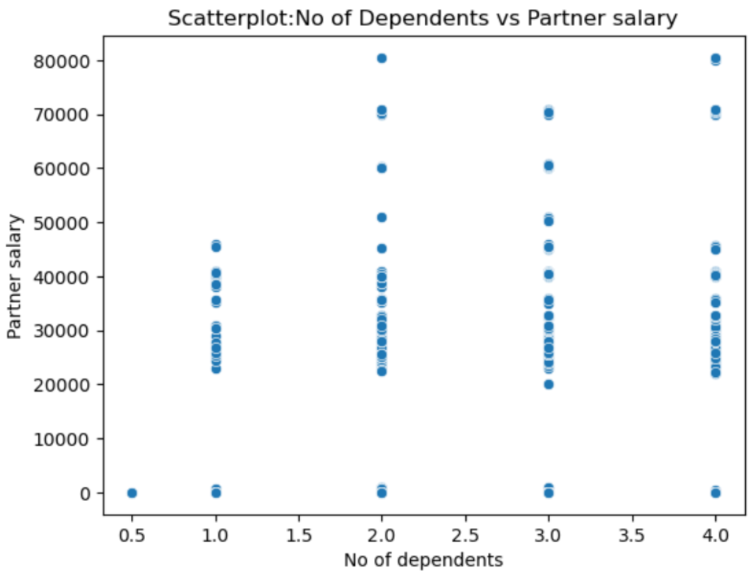
4.9.2.6 Is there a relationship between the number of dependents an individual has and their salary?



Observations:

- There is no correlation between number of dependents and salary.

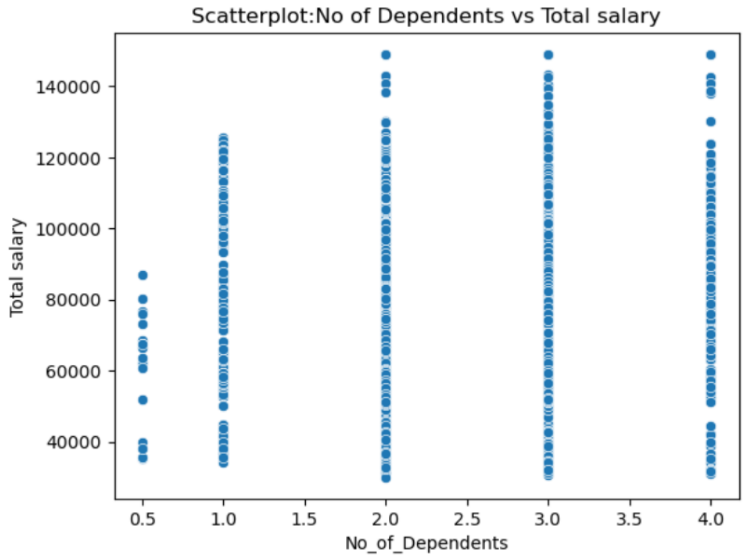
4.9.2.7 Is there any connection between the number of dependents an individual has and the salary of their partner?



Observations:

- There is no correlation between number of dependents and the salary of the partner.

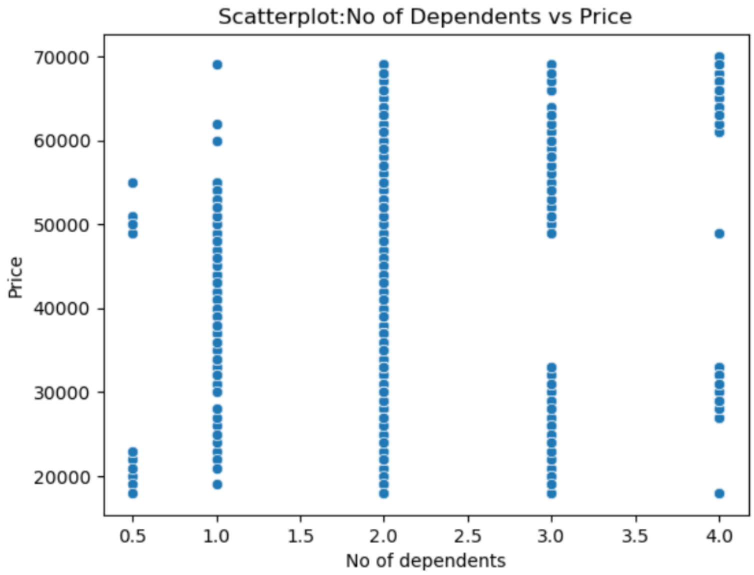
4.9.2.8 Can the number of dependents impact the overall family income?



Observations:

- There is no correlation between number of dependents and total salary of the family.

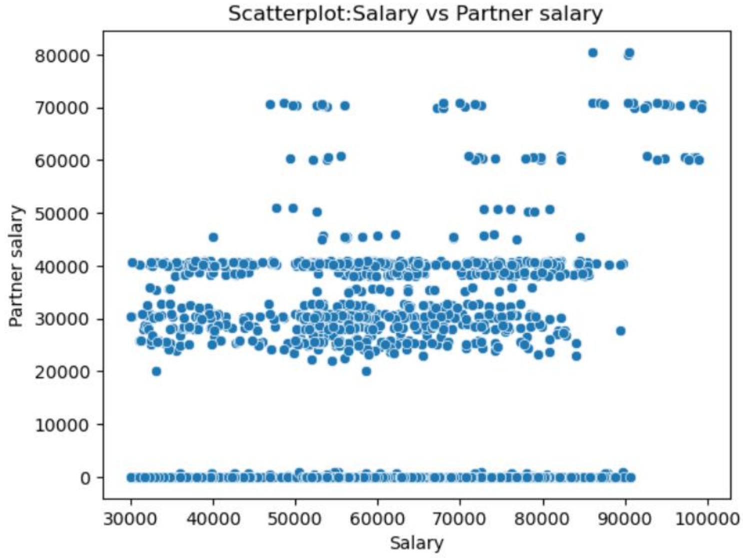
4.9.2.9 Could the number of dependents influence the price of the automobile an individual buys?



Observations:

- There is no correlation between number of dependents and the price of automobile.

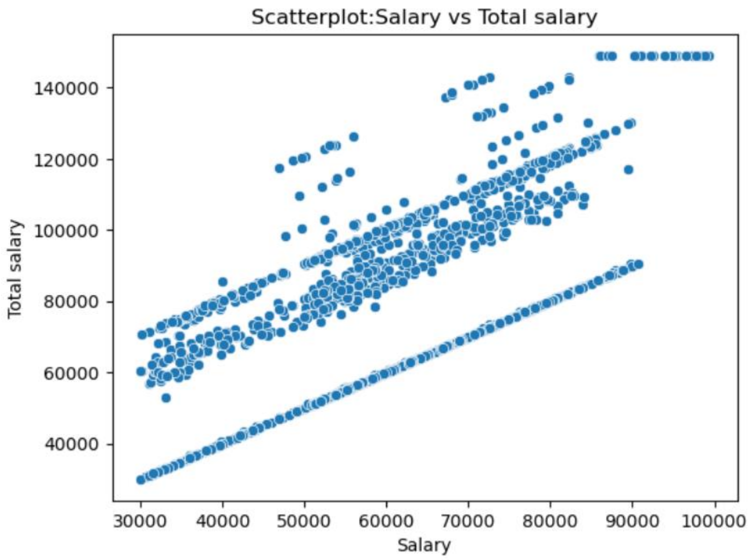
4.9.2.10 What is the correlation between an individual's salary and their partner's salary?



Observations:

- There is no much correlation between the salary of the customer and the salary of the partner, which is acceptable.

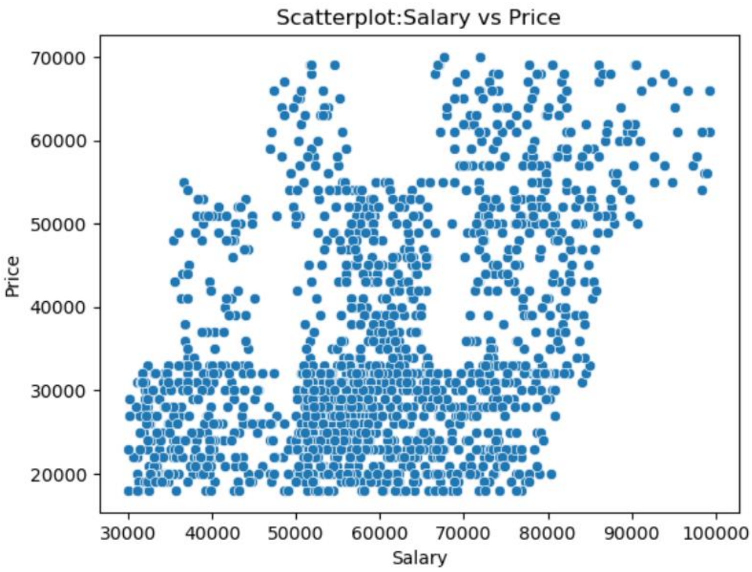
4.9.2.11 How much does an individual's salary contribute towards the overall family income?



Observations:

- There is a strong linear correlation, approximately 0.6, between salary and total salary, as salary constitutes major component of the total salary.

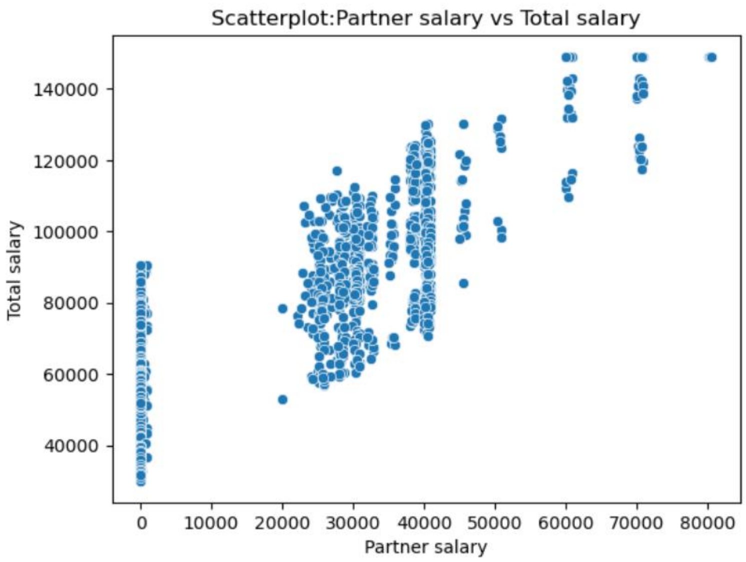
4.9.2.12 How does an individual's salary affect the price of automobile he buys?



Observations:

- Positive correlation about 0.4 is observed between salary and price. It implies that customers earning good tend to purchase automobile with higher price and comfort.

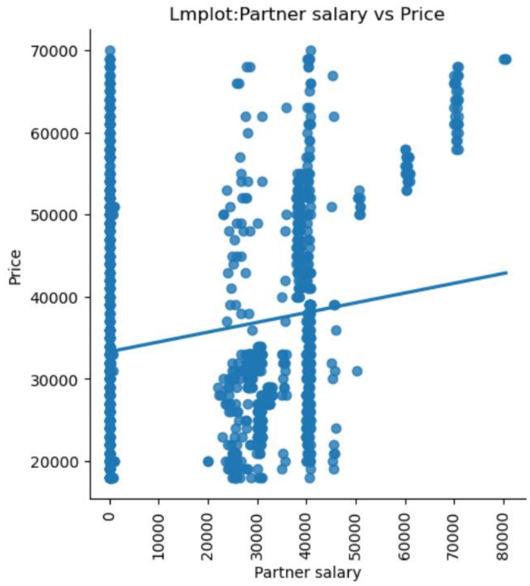
4.9.2.13 How much does the salary of a partner contribute towards the overall family income?



Observations:

- High correlation of 0.81 can be seen between salary and partner salary, as partner salary constitutes major component of the total salary.

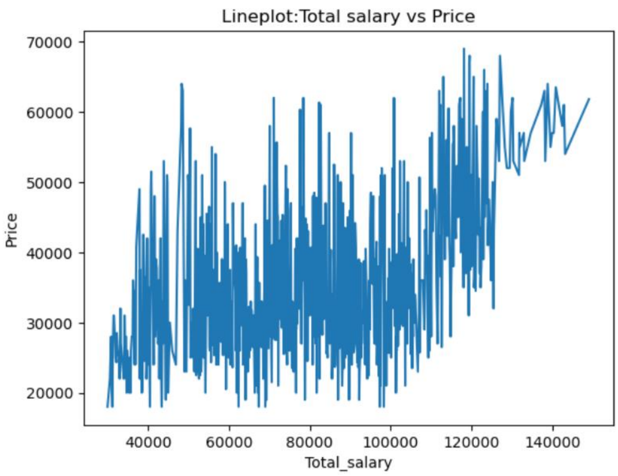
4.9.2.14 Does the second income influence the price of the automobile an individual buys?

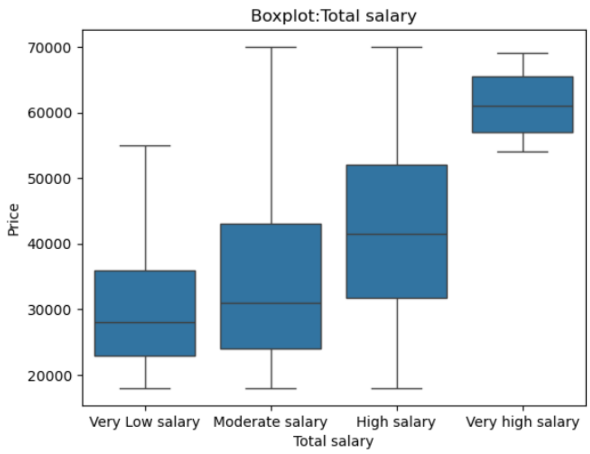


Observations:

- Positive correlation is seen between salary of the partner and price of the automobile purchased, as the second income generated helps in purchasing high end automobiles.

4.9.2.15 How does the total family income affect the price of the purchased automobile?





Observations:

- As the total salary increases, the price of the automobile purchased also increases.
- Half of the customers have seen to purchase automobiles ranging lesser than $40000.
- Automobiles ranging $60000-$70000 are purchased only by high earning individuals.

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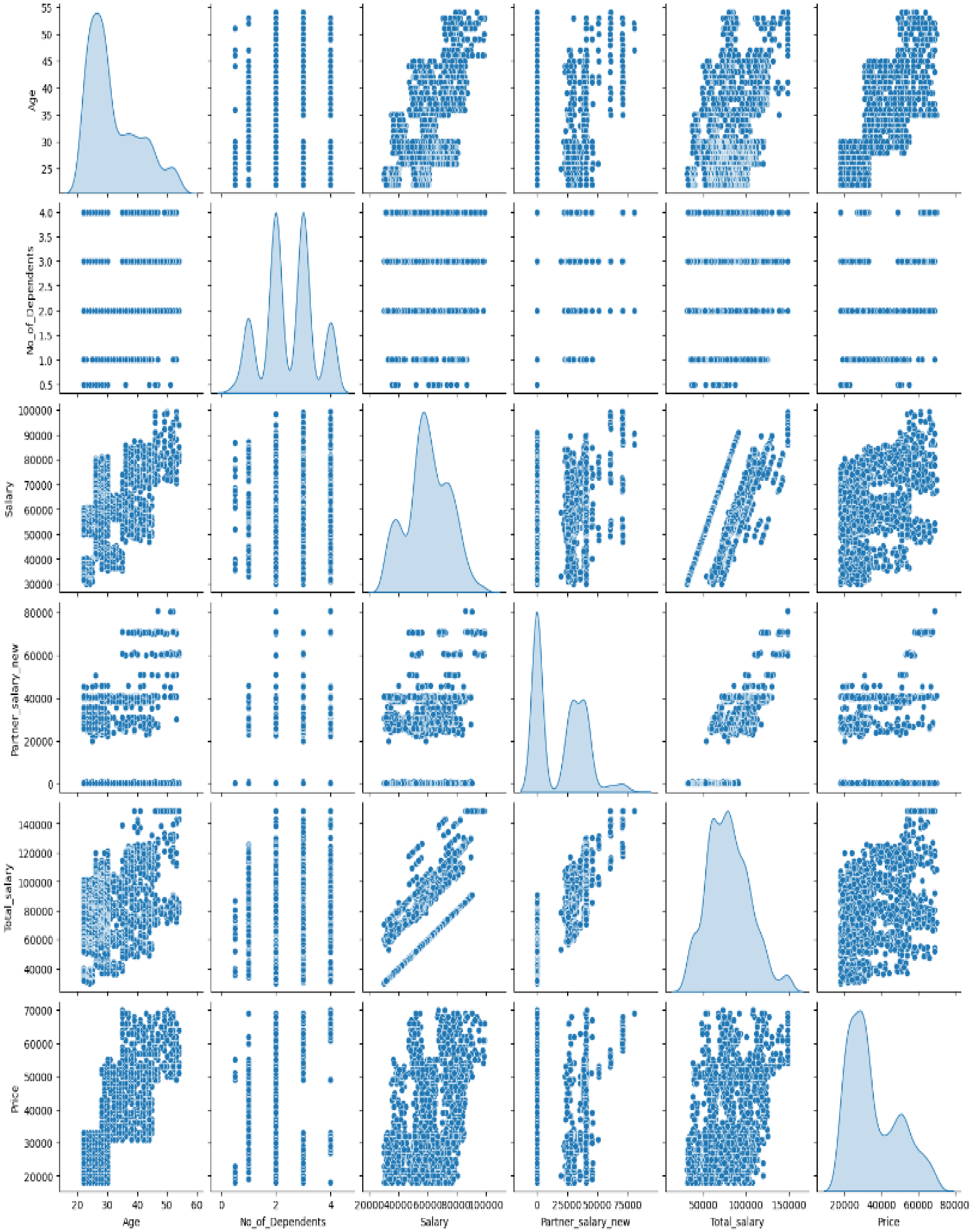
4.9.2.16 Pair plot to describe relationship between all numerical variables

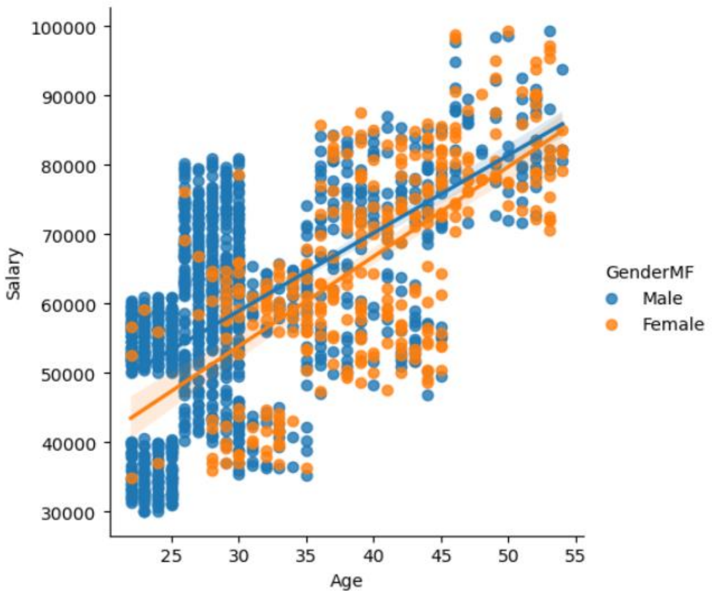


4.9.3 Insights of Relationship Between Numerical Variables

- Typically, the decision to buy a car is influenced by the family's income, their transportation needs and the number of dependents in the family.
- When individuals have high salary, their partner's salary is substantial and the total family income is elevated, they are more inclined to purchase high-value cars. This falls in line with the analysis conducted.
- While high priced cars can accommodate more people, negative correlation is observed between the number of dependents and the price of the cars purchased. The numbers of dependents in a family do not directly influence the price of the automobile or the income of the family. However, almost 98% of the families have dependents which relates to the need of a four-wheeler in a family.
- As experience increases with age, the income generated by a family also increases. So older people tend to purchase high-value cars. As cars have been sold in various price ranges, a significant portion of automobile purchases is made by individuals under the age of 30.

4.9.4 Relationship between categorical vs numerical variables

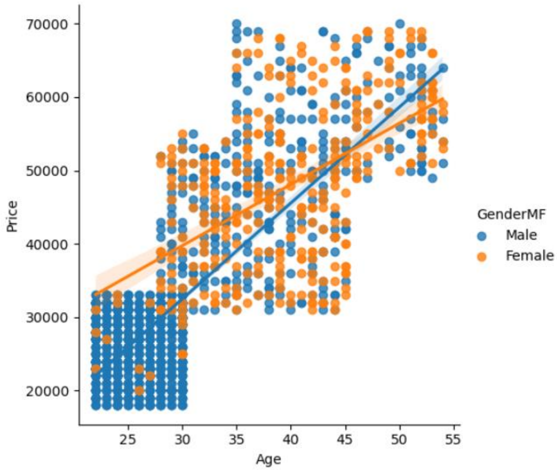
4.9.4.1 Given that age is a significant factor influencing salary, how does it differ between genders?



Observations:

- In case of both male and female customers, the relationship between age and salary is linear. As age increases, salary also increases.
- Between 25-40 years of age, the male customers earn higher salary than the female customers.
- Above 50 years of age, the salary almost matches.

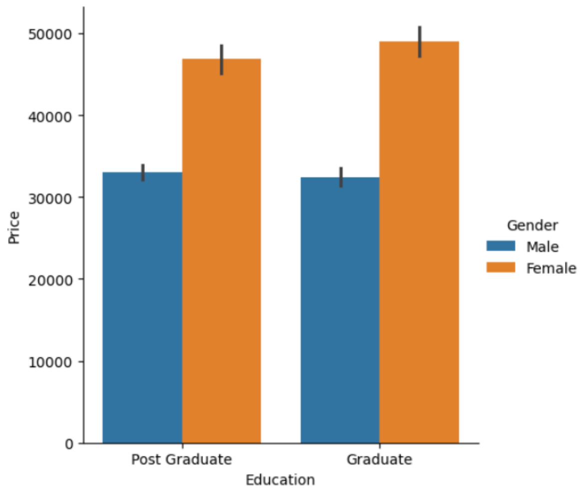
4.9.4.2 Considering that age plays a crucial role price of influencing the price of automobiles, how does it differ between genders?



Observations:

- In case of both male and female customers, the relationship between age and price of the automobile is linear. As age increases, the price of the automobile considered also increases.
- Men below 30 years purchase significant number of cars priced between $20000-$35000.
- Women are more likely to buy high-priced cars compared to men.

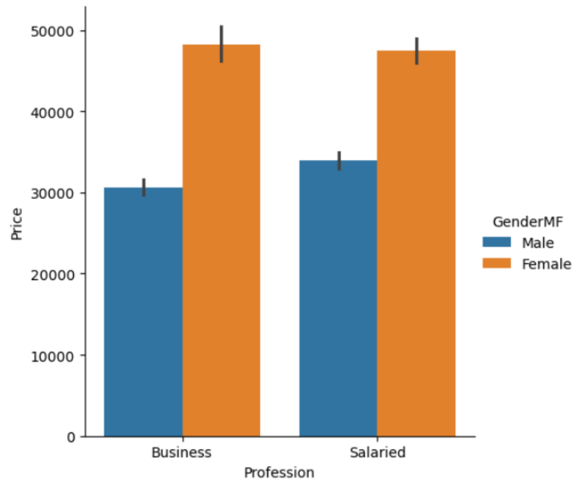
4.9.4.3 What impact does education have on the price of the automobile purchased?



Observations:

- The pattern seen among graduates and postgraduates for spending on automobiles is the same.
- Both graduate and postgraduate female customers spend more on automobiles than the male customers.
- Female graduates are seen to spend a little higher on automobiles.
- Education may not be a parameter to be considered on purchase of automobiles.

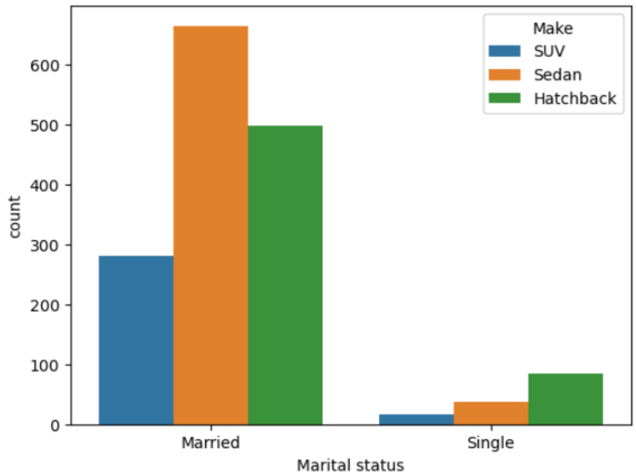
4.9.4.4 How does one's profession affect the price of the automobile purchased?



Observations:

- There is no much difference between salaried individuals and individuals involved in business on the amount spent of automobile purchase.
- Both spend almost the same amount on automobile purchase.

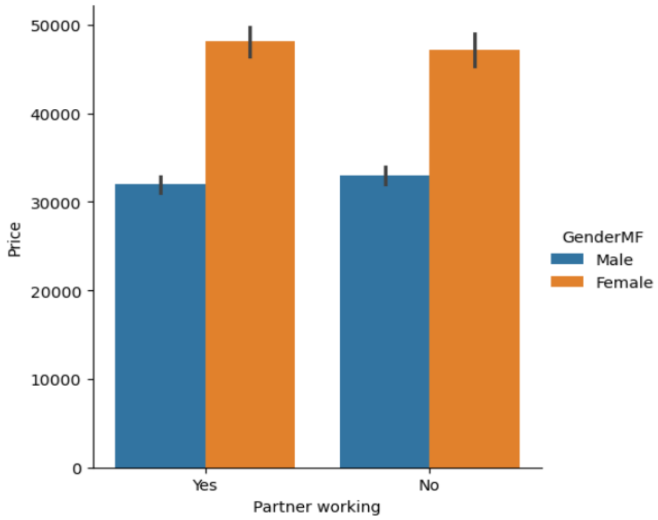
4.9.4.5 What is the correlation between the marital status and the choice of car brand?

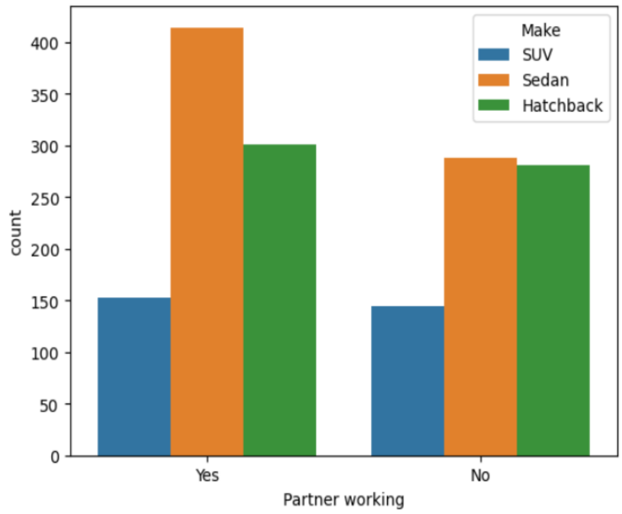


Observations:

- Married individuals buy more cars regardless of the brand.

4.9.4.6 Does the presence of a working partner influence the price based on gender and selection of car brand?

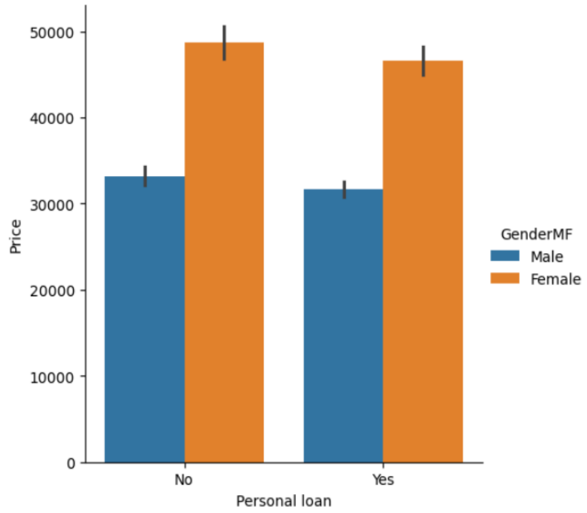


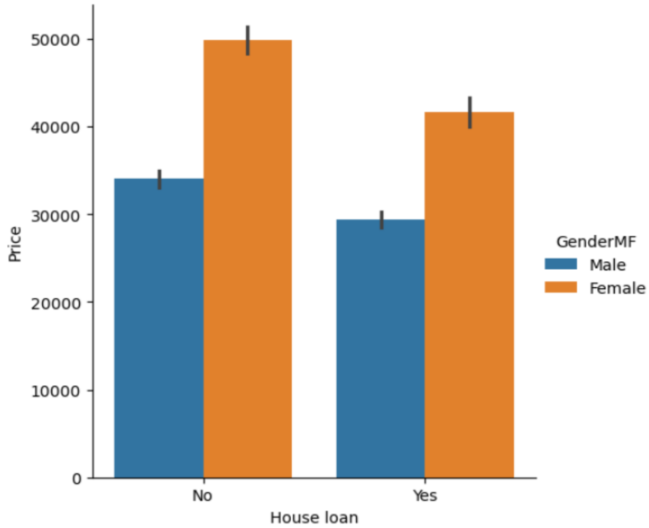


Observations:

- Having a partner who is employed does not affect the price of the automobile purchased.
- However, with working partners, the percentage of sedans are much higher.

4.9.4.7 Does the presence of loan influence the price of the automobile purchased based on gender?

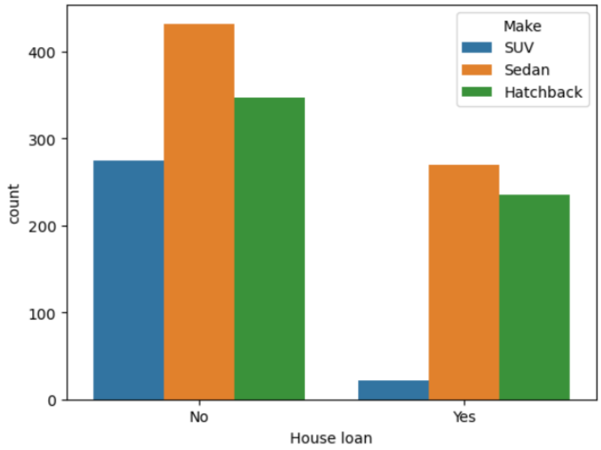


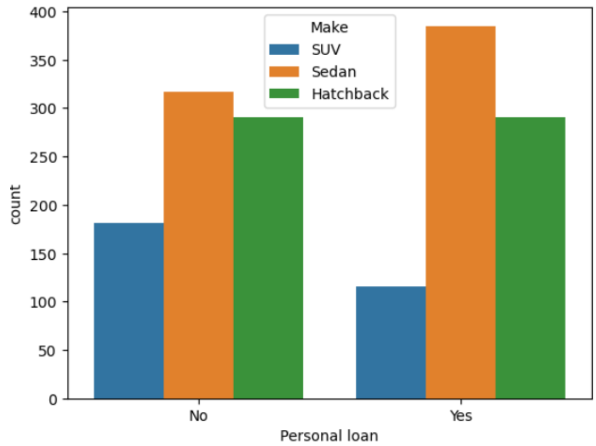


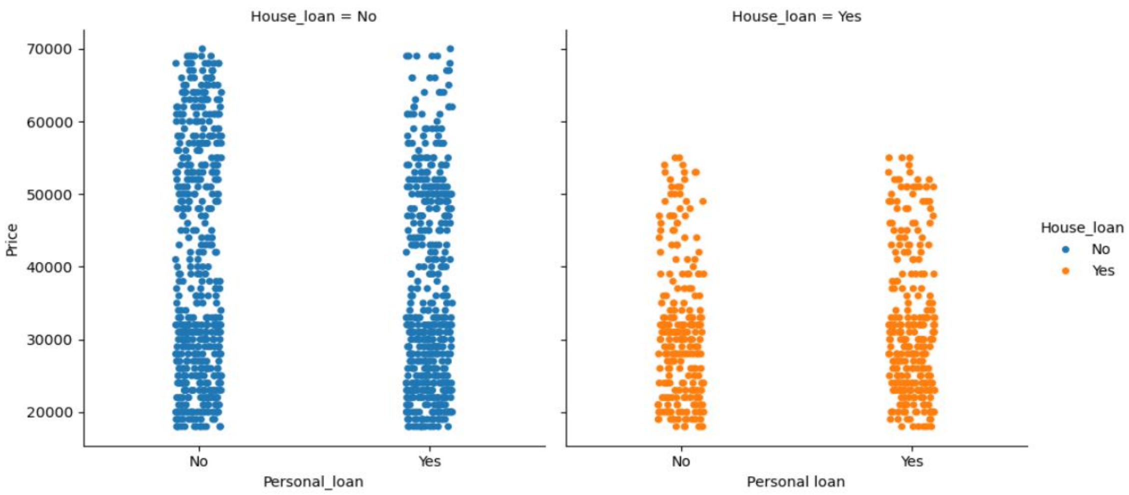
Observations:

- Individuals having personal loan spend lesser on automobiles.
- Individuals having house loan spend lesser on automobiles.

4.9.4.8 What is the impact of loans on the selection of car brand?







Observations:

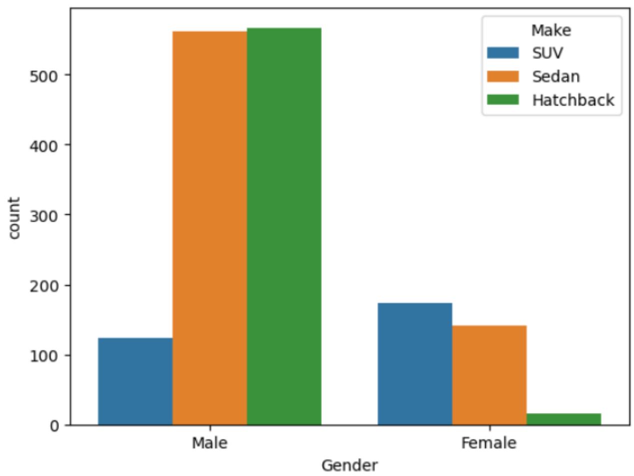
- Individuals without house loans or personal loans or both loans tend to buy automobiles.
- Individuals without loans, prefer buying SUVs by 20%, followed by Sedan.
- The number of SUV purchased by customers having both the loans is lesser only around 11, followed by hatchback (116) and Sedan (151).

4.9.5 Insights of Relationship Between Numerical Variables and Categorical Variables

- Men between age 25-30 exhibit higher inclination towards purchasing cars, whereas women over age 30 prefer cars. When targeting customers of different genders, age serves as a primary criterion.
- Both men and women with substantial salaries prefer high priced cars with increase in salary.
- People who are not burdened by loans are more inclined to choose to purchase cars. The percentage of SUVs is very low with people having loans. They tend to choose sedans over SUVs as SUVs are highly priced.
- Married individuals have a tendency to buy cars more frequently.
- Education and profession might not hold significant weight as criteria when purchasing cars. This could be attributed to the fact that not all employed individuals work in the same field they studied.
- Sedans are highly sold. The comfort and the pricing available in the model might be a factor influencing the choice of car. As the number of dependents are on an average 2-3, sedans are much suitable and comfortable.

5. Key Questions

1. Do men tend to prefer SUVs more compared to women?

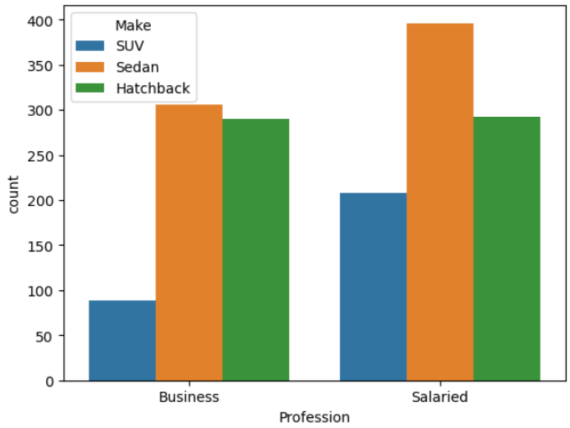


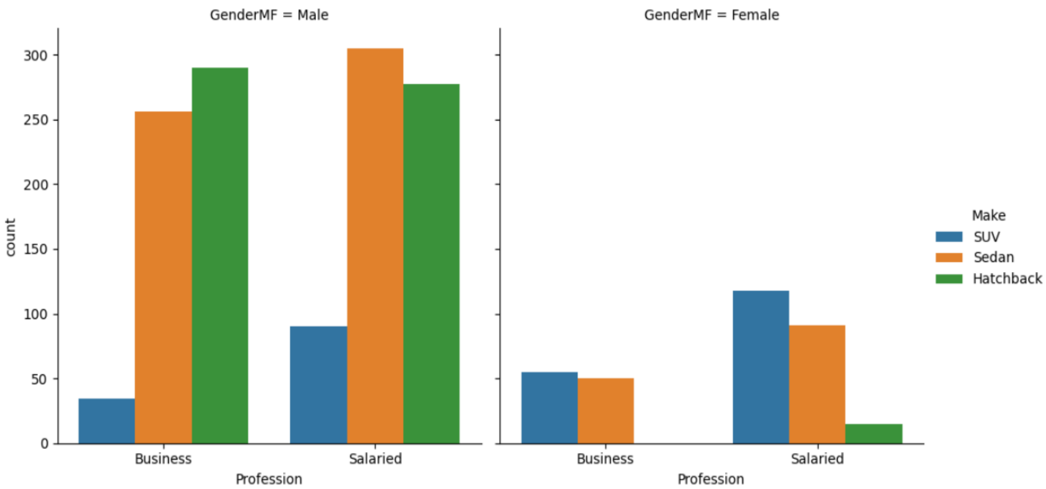
Gender	Make	Count
Female	SUV	173
	Sedan	141
	Hatchback	15
Male	SUV	118
	Sedan	156
	Hatchback	565

Answer / Observation:

- No. Women exhibit a preference for SUVs nearly 20% higher than the men do.

2. What is the likelihood of a salaried person buying a Sedan?

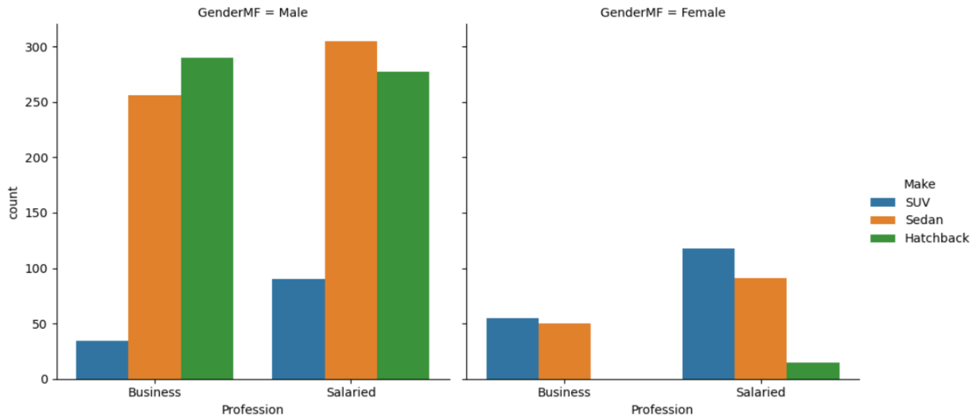




Answer / Observation:

- Salaried customers show a preference for purchasing sedans over other models. This preference for sedans is 14% greater than that for hatchbacks and 19% greater than that for SUVs.
- Among the salaried customers, salaried men are likely to prefer sedans by 45% and salaried women are likely to prefer SUVs by 52%.

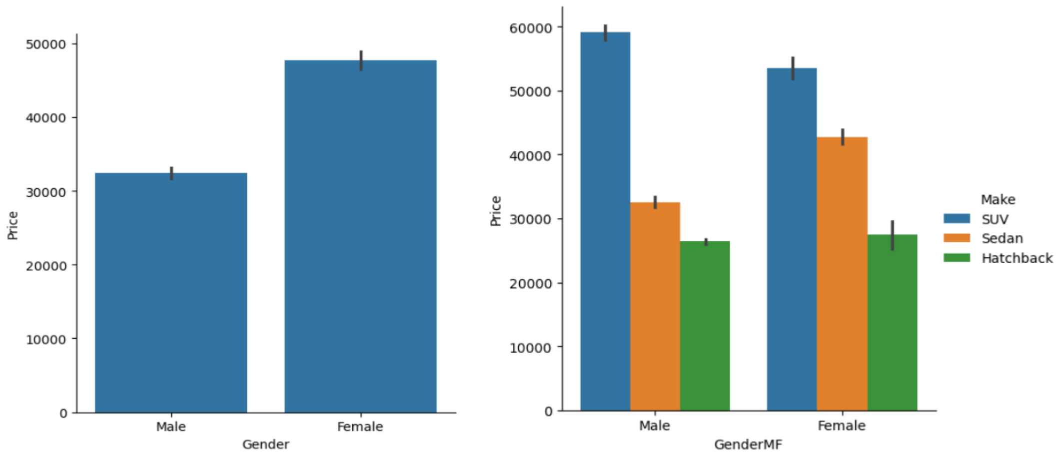
3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?



Answer / Observation:

- No, Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale is false. Salaried women are the easier targets for SUV. And Salaried men are an easier target for Sedan.

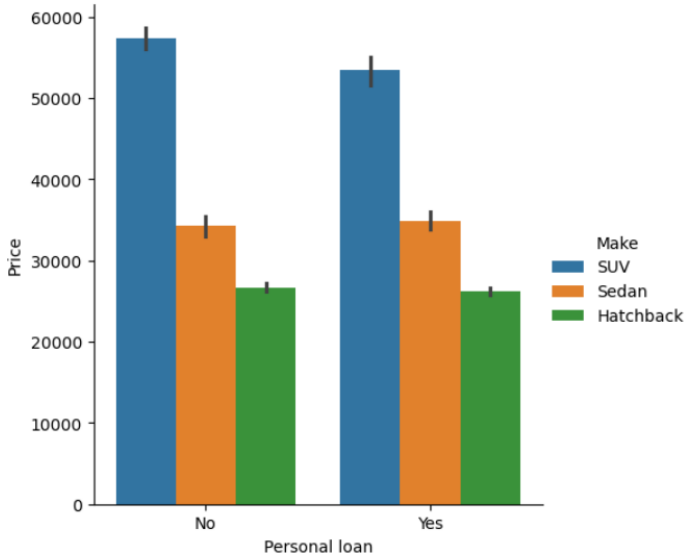
4. How does the amount spent on purchasing automobiles vary by gender?

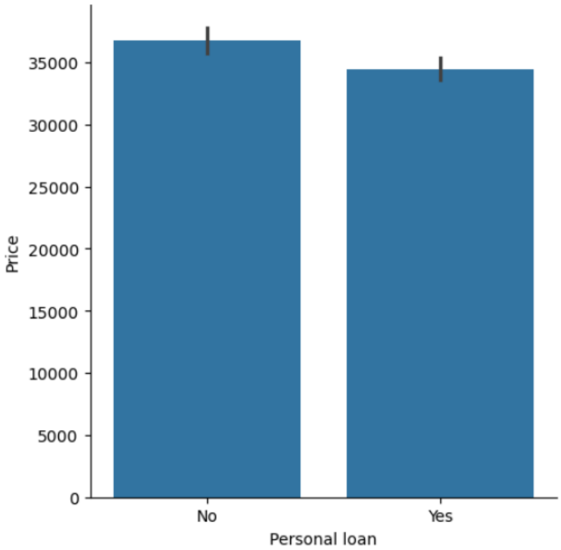


Answer / Observation:

- On an average, the amount spent on purchasing automobiles is higher by women compared to men. High priced cars are preferred by women over men.
- In comparison to SUVs, women tend to allocate more spending towards hatchbacks and sedans.
- Females spend more on purchasing automobiles than men. On an average, Women spend 47705 dollars on automobile than men who spend 32817 dollars.
- Though women have higher tendency towards SUVs, the variants chosen are low-priced ones. Hence men spend more on SUV, followed by sedan and hatchback compared to women.

5. How much money was spent on purchasing automobiles by individuals who took a personal loan?



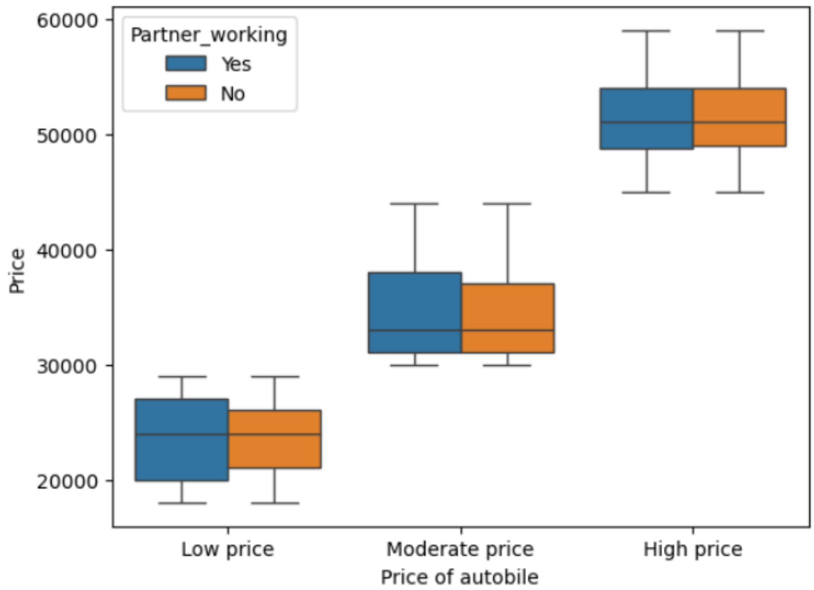


Personal loan	Gender	Average price
No	Female	48000
	Male	33000
Yes	Female	46000
	Male	32000

Answer / Observation:

- The average amount of money spent on purchasing automobiles by individuals who took a personal loan is slightly higher than the individuals without personal loans.

6. How does having a working partner influence the purchase of higher-priced cars?



Make	Gender	Partner working	Average price
Hatchback	Female	No	28700
		Yes	26375
	Male	No	26262
		Yes	26621
SUV	Female	No	53375
		Yes	53569
	Male	No	59671
		Yes	58483
Sedan	Female	No	41546
		Yes	43792
	Male	No	33584
		Yes	31862

Answer / Observation:

- Having a working partner has no much influence on the purchase of automobiles of all price range.

6. Conclusions

In contemporary society, buying a car has evolved into a symbol of status. Aligned with this assertion, we see individuals across all salary ranges purchasing cars. However, men tend to dominate in number for purchasing cars.

The criteria for purchasing cars would be the need of the individual, number of dependents, budget and brand. Majority of people buy cars which are sold in the mid-range of $45000, which suggests that, these are the target for individuals earning in a moderate range.

Though men tend to dominate in number, we can see women aligning towards high priced cars.

7. Actionable Insights and Business Recommendations

- Our target demographic for automobile sales includes men aged between 25-30 and women above aged over 30.
- Target customers who are free from mortgage or personal loans, or alternatively, consider marketing lower-priced variants to those with existing loans.
- Considering married men or women as the target audience is a crucial parameter in car marketing.
- It is essential to remember that men typically gravitate towards sedan followed by hatchback, whereas women tend to prefer SUVs.
- We should target families with high total income, which can serve as a criterion for selling high-end variants across all three models.
- Since the majority of the customers have dependents, targeting individuals with dependents in the family appears to be a more promising approach.
- Taking customer feedback into account would be beneficial for further analysis aimed at increasing sales, especially considering the parameter that SUVs and hatchbacks have lower sales compared to sedans.